

Microservices and the API Gateway

Microservices is an architectural style for building applications that are structured into several loosely coupled services. An API gateway acts as a single entry point into the system. This article, aimed at developers who work in API, mobile and Web development, explains how the two can be used in tandem.



An API gateway is a management tool to create, publish, maintain, monitor and secure APIs. You can create APIs that carry out a specific function of your logic or that can access other Web services.

To understand this better, consider the following example. We are building a client for a shopping application and need to implement a 'Product details' page, which displays information about any given product.

Figures 1(a), 1(b) and 1(c) show what you will see when scrolling through the 'Product details' page in any shopping site (this one is from Amazon). The page also shows other information, which includes:

1. Selected product information
2. Order history
3. Cart
4. Reviews of the product
5. Inventory and shipping
6. Various recommendations, including other products frequently purchased with this product, and so on.

Monolithic architecture

The program (in any language or framework) for this can be developed using a monolithic architecture. This is nothing but a traditional unified model for the design of a software program. In this context, monolithic means all the components are interconnected and interdependent, rather than loosely coupled. In this tightly coupled architecture, all